

Léo Dana

PhD on the mathematics of AI, Inria Paris

Paris, France

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26 years old, [website](#)



Jobs

2025-2028 **PhD, Inria, Paris**



I am currently a PhD at Inria Paris, under the supervision of Francis Bach and Loucas Pillaud-Vivien. My work revolves around understanding neural network from multiple view-points: optimization or mechanistic.

Education

2020-2025 **Bachelor and Master of Mathematics, École Normale Supérieure, Lyon**



Courses in geometry, topology, algebra and probability, with additional computer science courses. Specialization in Probability, Statistics and Machine Learning

2018-2020 **Preparatory Classes, Lycée Janson de Sailly, Paris 16-th**

Preparation for a highly competitive examination to enter scientific universities. Undergraduate courses in mathematics, physics and computer science. Graded 17th at *ENS de Lyon*.

Internships

Spring 2025 **Research, Kayrros, Paris**



A 22-week internship on developing deep-learning algorithms to detect methane leaks using geostationary satellite images.

Fall 2024 **Research, Inria, Paris**



A 22-week internship directed by Francis Bach and Loucas Pillaud-Vivien on the convergence of shallow ReLU networks to global minimum of the loss. (Article available on arXiv [here](#).)

Spring 2024 **Research, Dauphine - Lamsade, Paris**



A 18-week internship on Mechanistic Interpretability to understand memorization in deep-learning models directed by Yann Chevalyre. (Article published at AISTATS [here](#).)

Summer 2023 **Research, FAR AI & SERI MATS, Berkeley, US**



A 20-week internship on Mechanistic Interpretability and activation vectors, building a gear-level understanding of Large Language Model's behavior, directed by Claudia Shi.

Summer 2022 **Research, IRT Saint Exupéry, Toulouse**



A 10-week internship on Conformal Prediction and large deviation inequalities, directed by Sébastien Gerchinovitz.

Summer 2021 **Research, I2M (Institut de Mathématique de Marseille), Marseille**

A 6-week internship on reinforced random walks modeling ant behavior, directed by Bruno Schapira.

Work and Volunteering

2022-2025 **Co-Founder**, [EffiSciences](#)



I helped create the association EffiSciences, whose goal is to promote impactful research, and I have co-written a report on how to make French research more impactful.

2022 **AI Safety Volunteer**, [ML4Good](#)



ML4Good

I co-organized the first two ML4Good bootcamps, which is a 20-person week-long summer boot camps on AI safety.

2021-Present **Seminars**

I taught or helped in various seminars : [Category Theory](#) (ENS de Lyon, 2021), [Fractal Geometry](#) (ENS de Lyon, 2022), [Turing Seminar](#) (MVA master at ENS Paris-Saclay in 2024, 2025 and 2026), [General-Purpose AI Policy Lab](#) training course (Science-Po, 2025)

2021-2022 **Oral exam preparator**, *Les Lazaristes*, Lyon

Dispensing math oral exam to the students preparing for nationwide competitive exams in engineering

Skills and interests

Languages : French (mother tongue), English (C1)

IT : Latex, General Python, Pytorch, [GitHub](#)

Sports : Cycling, Running, Strength training, Brazilian Jui-jitsu